

# Business Requirements Document

## AI Travel Planner

AI-powered travel planning, conversational assistance, email communication and customer feedback

|                   |                                                            |
|-------------------|------------------------------------------------------------|
| Document Version  | 1.0                                                        |
| Document Type     | Business Requirements Document (BRD)                       |
| Project Type      | Web Application / SDLC Demonstration                       |
| Primary Users     | Travelers, Administrators, Travel Consultants              |
| Core Capabilities | AI Planner, AI Chat, Email, Itinerary Management, Feedback |

## 1. Business Objective

The objective is to develop an AI-powered Travel Planner that allows users to describe their travel requirements and receive a personalized, editable itinerary.

The application will combine a structured trip-planning form, an AI planner, a conversational chat box, email communication, itinerary management, and customer feedback.

The product is intentionally designed to support an evolving SDLC demonstration where requirements can change after customer reviews and each change can be implemented and released as a new iteration.

## 2. Business Problem

Travel planning normally requires users to research destinations, activities, restaurants, accommodation, transportation and costs across multiple sources.

The proposed system provides a single application where users can enter their preferences, generate an initial travel plan, discuss changes with an AI assistant, save the result, and receive the itinerary by email.

## 3. Project Goals

Provide a simple and engaging travel-planning experience.

Use AI to generate and personalize itineraries.

Provide a conversational chat interface for refining an itinerary.

Allow users to save, edit, regenerate and share plans.

Send itinerary and notification emails.

Capture customer feedback and use it as input for future product iterations.

Provide an administrator view for managing users, destinations, feedback and application metrics.

## 4. Target Users

**Traveler:** creates trips, interacts with the AI planner, manages itineraries and provides feedback.

**Administrator:** manages users, destinations, feedback, content and system-level reporting.

**Travel Consultant (optional):** reviews or enhances AI-generated plans for customers.

## 5. Core User Journey

Register/Login → Create Trip → Enter Requirements → Generate AI Plan → Review Itinerary → Chat with AI → Modify/Regenerate → Save Trip → Email/Share → Travel → Provide Feedback.

## 6. Trip Creation

Users can create a trip by entering trip name, destination, start date, end date, number of travelers, adults, children, budget and currency.

Example: Tokyo Family Holiday, Tokyo, 10–17 October 2026, 4 travelers, USD 5,000.

## 7. Travel Preferences

Travel style: Relaxed, Balanced, Adventure, Luxury, Budget, Family, Business or Cultural.

Interests: History, Nature, Shopping, Food, Museums, Beaches, Nightlife, Photography, Adventure, Sports, Local Culture and Architecture.

Food preferences: No Preference, Vegetarian, Vegan, Halal, Gluten-Free or Other.

Transportation: Public Transport, Taxi, Rental Car, Walking or Mixed.

## **8. AI Travel Planner**

The AI Planner is the primary intelligence feature. It receives structured trip requirements and generates a day-by-day itinerary.

AI output should include suggested activities, times, durations, estimated costs, locations and reasons for recommendations.

The generated plan should be editable rather than treated as a fixed result.

## **9. AI Chat Box / Conversational Assistant**

The application must provide a chat box associated with the current trip.

Users can ask natural-language questions or request changes such as: "Make Day 2 less busy", "Remove shopping", "Replace museums with outdoor activities", or "Make this suitable for children."

The AI should use the current itinerary and trip preferences as context.

Changes should be previewed or clearly reflected in the itinerary so the user can review them before relying on the updated plan.

## **10. AI Itinerary Regeneration**

Users should be able to regenerate a complete itinerary or a specific day/section.

Small changes should not require unnecessary regeneration of unrelated parts of the trip.

The system should retain the user's selected preferences when regenerating.

## **11. Budget Estimation**

The system should estimate costs for accommodation, food, transportation, activities, shopping and other expenses.

It should show estimated total cost versus the user's budget.

Users may ask the AI to reduce the estimated cost or suggest lower-cost alternatives.

## **12. Accommodation and Activity Preferences**

Users can specify accommodation type, budget range, preferred location, rating and facilities.

Activities should support view, edit, remove, replace and move-to-another-day operations.

## **13. Saved Trips**

Users can save generated trips and reopen them later.

Users can edit, regenerate, delete and share saved itineraries.

## **14. Email Communication**

Email is a core feature for the demo.

After itinerary generation, the user can receive an itinerary email.

Users can receive notifications when a significant itinerary change is made.

The system can send trip reminders before the travel date.

Users can share an itinerary by entering another recipient's email address.

Example email events: Trip Created, Itinerary Updated, Trip Reminder and Itinerary Shared.

## 15. Customer Feedback

Users should be able to rate the generated itinerary and provide comments.

Feedback should be stored against the relevant trip/itinerary.

Administrators should be able to review feedback and identify recurring requirements.

## 16. AI Feedback Analysis

AI may summarize customer feedback and identify recurring themes.

Example themes: schedules are too busy, recommendations are too expensive, insufficient family activities, or insufficient transportation information.

This feature can be introduced as a later SDLC iteration.

## 17. User Account Management

Users can register, log in, log out and update their profile.

Profile preferences may include preferred currency, default travel style and dietary preference.

## 18. Admin Dashboard

Dashboard metrics may include total users, total trips, generated itineraries, popular destinations, average budget, feedback volume and AI usage.

Administrators can manage users, destinations and feedback.

## 19. Destination Management

Administrators can add, edit, disable or remove destinations.

Destination information can include description, popular activities, recommended duration and travel information.

## 20. Search and Filtering

Users can search saved trips and destinations.

Filters may include destination, country, city, budget, travel style and trip duration.

## 21. Notifications

The system should provide in-app feedback for successful or failed operations.

Email notifications should be configurable for key events.

AI-generated information should be clearly identified as recommendations rather than confirmed bookings.

## 22. Non-Functional Requirements

**Security:** protect accounts, passwords, APIs and personal information.

**Performance:** normal application pages should respond within an acceptable time under expected usage.

**Reliability:** saved trips must persist across application restarts.

**Maintainability:** separate UI, business logic, database, AI integration and email services.

**Scalability:** architecture should allow additional AI features and users later.

## 23. Suggested High-Level Architecture

Browser/UI → Web API → Business Services → Database.

AI Service integrates with the application through a dedicated service layer.

Email Service handles transactional and notification emails.

Separating AI and email integrations makes future provider changes easier.

## 24. Key Functional Modules

Authentication & User Management

Trip Management

Travel Preference Management

AI Planner

AI Chat Box

Itinerary Management

Budget Estimation

Email Notification Service

Feedback Management

AI Feedback Analysis

Destination Management

Admin Dashboard

## 25. SDLC Demonstration Roadmap

**Release 1 – MVP:** login, create trip, destination, dates, budget, basic itinerary and save trip.

**Customer Feedback:** “The itinerary is too generic.”

**Release 2:** interests, travel style, dietary requirements and AI personalization.

**Customer Feedback:** “I want to change the itinerary by chatting instead of editing everything manually.”

**Release 3:** AI chat box and conversational itinerary modification.

**Customer Feedback:** “I need to know whether the plan fits my budget.”

**Release 4:** cost estimation and AI budget optimization.

**Customer Feedback:** “Send the final itinerary to my email.”

**Release 5:** itinerary email, sharing and reminders.

**Customer Feedback:** “Management needs to understand what customers are asking for.”

**Release 6:** feedback dashboard and AI feedback analysis.

## 26. Example Customer Change Requests

“Can you make the trip more suitable for children?”

“Can you reduce the number of activities per day?”

“Replace expensive restaurants with budget options.”

“Send the final itinerary to my email.”

“Can I ask the AI questions about my trip?”

“Show me why the AI recommended this activity.”

“Can management see the most common customer complaints?”

## 27. Demo Acceptance Criteria

The user can create an account and create a trip.

The user can enter destination, dates, travelers, budget and preferences.

The AI can generate a structured itinerary.

The user can interact with the itinerary through a chat box.

The AI can make requested changes while retaining trip context.

The user can save and reopen a trip.

The user can receive the itinerary through email.

The user can submit feedback.

The administrator can view trips and feedback.

The demo can show at least two requirement changes resulting from customer feedback.

## 28. Future Enhancements

Live weather integration.

Real-time flight and hotel information.

Map integration.

Calendar integration.

Multi-language itinerary generation.

Voice-based AI travel assistant.

Mobile application.

AI-based travel risk and disruption notifications.

Integration with booking providers.

## 29. Important AI Considerations

AI-generated recommendations should be presented as recommendations and not as guaranteed availability, prices or bookings.

Users should be able to review AI-generated changes.

Sensitive personal information should not be unnecessarily included in AI prompts.

AI failures should have a clear fallback message rather than blocking the entire application.

AI prompts and outputs should be logged only according to the application's privacy and retention requirements.

## 30. High-Level Requirements Matrix

| ID    | Requirement                       | Priority      |
|-------|-----------------------------------|---------------|
| FR-01 | User registration and login       | High          |
| FR-02 | Create and manage trips           | High          |
| FR-03 | Enter travel preferences          | High          |
| FR-04 | AI itinerary generation           | High          |
| FR-05 | AI conversational chat box        | High          |
| FR-06 | Modify/regenerate itinerary       | High          |
| FR-07 | Budget estimation                 | Medium        |
| FR-08 | Save and reopen trips             | High          |
| FR-09 | Email itinerary and notifications | High          |
| FR-10 | Customer feedback                 | Medium        |
| FR-11 | Admin dashboard                   | Medium        |
| FR-12 | AI feedback analysis              | Future/Medium |

End of Business Requirements Document